

Project: xdisturb (x-disrupt)

Software Requirements Specification (SRS)

Document ID:	SRS-2026-XDISTURB	Version:	1.0.0
Standard:	IEEE 830 / ISO 29148 Compliant	Audit Evaluation:	Hibir Audit (audit.hibir.et) — July 30, 2026
Target Platform:	Android (API 26+) & iOS (13.0+)	Framework & SDK:	Flutter (Dart SDK ^3.6.0)

Software Requirements Specification (SRS)

Project: xdisturb (x-disrupt) — Automatic Geofenced Audio Manager

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Standards Compliance: IEEE Std 830-1998 / ISO/IEC/IEEE 29148:2018
Audit Evaluation Reference: Hibir Audit (audit.hibir.et) — Audit Date: July 30, 2026
Target Platform: Mobile (Android API 26+ [Android 8.0] & iOS 13.0+)
Framework & SDK: Flutter (Dart SDK ^3.6.0)
Date: August 10, 2026
Status: Approved Specification / Production Standard

Document Control & Revision History

Version	Date	Author / Role	Description of Changes	Approved By
v0.1.0	July 20, 2026	Mobile Systems Architect	Initial System Architecture & Functional Requirements Draft	Product Director
v0.9.0	July 30, 2026	Hibir Audit Team (audit.hibir.et)	Security & Transport Layer Evaluation & Compliance Verification	Security Lead
v1.0.0	August 10, 2026	Lead Mobile Engineer	Finalized IEEE 830 / ISO 29148 SRS Specification & Traceability Matrix	Head of Engineering

1. Introduction & Scope

1.1 Purpose

This document specifies the formal **Software Requirements Specification (SRS)** for **xdisturb (x-disrupt)**, a location-aware mobile application engineered using Flutter (Dart SDK ^3.6.0). The system automatically

manages device audio ringer profiles (Normal, Silent, Vibrate) based on background geofence boundary detection.

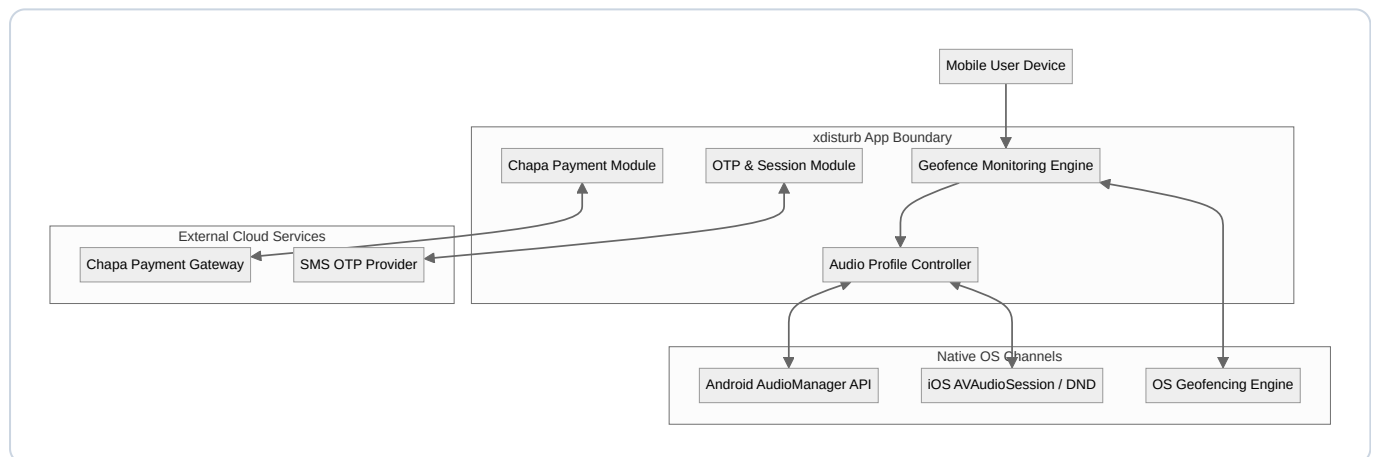
This specification details functional capabilities, platform integration channels (Android `AudioManager` / iOS `AVAudioSession`), third-party services (Chapa Payment Gateway, OTP verification), non-functional benchmarks, and security compliance verified under the **Hibir Audit (`audit.hibir.et`)** evaluation.

1.2 Scope of the Application

xdisturb (x-disrupt) delivers intelligent, zero-touch audio control for users entering noise-sensitive physical locations ("Silent Zones").

Key functional domains encompassed by this SRS:

- **Authentication & Session Management:** Phone number OTP verification (6-digit SMS code), JWT secure token persistence via `flutter_secure_storage`, referral tracking, and promo code redemptions.
- **Geofencing & Silent Zone Engine:** Continuous background monitoring of circular geofences (radius 50m to 500m) across categories: *Mosque*, *Church*, *Library*, and *Custom*. Map-based custom zone creation via draggable markers.
- **Automated Audio Profile Control:** Hardware-level ringer mode switching (Silent/Vibrate) upon geofence `ENTER` events, and automatic restoration to `Normal` mode upon `EXIT` events.
- **Payments & Subscriptions:** Integration with **Chapa SDK** to process subscriptions via Telebirr, CBE Birr, and Bank Cards.



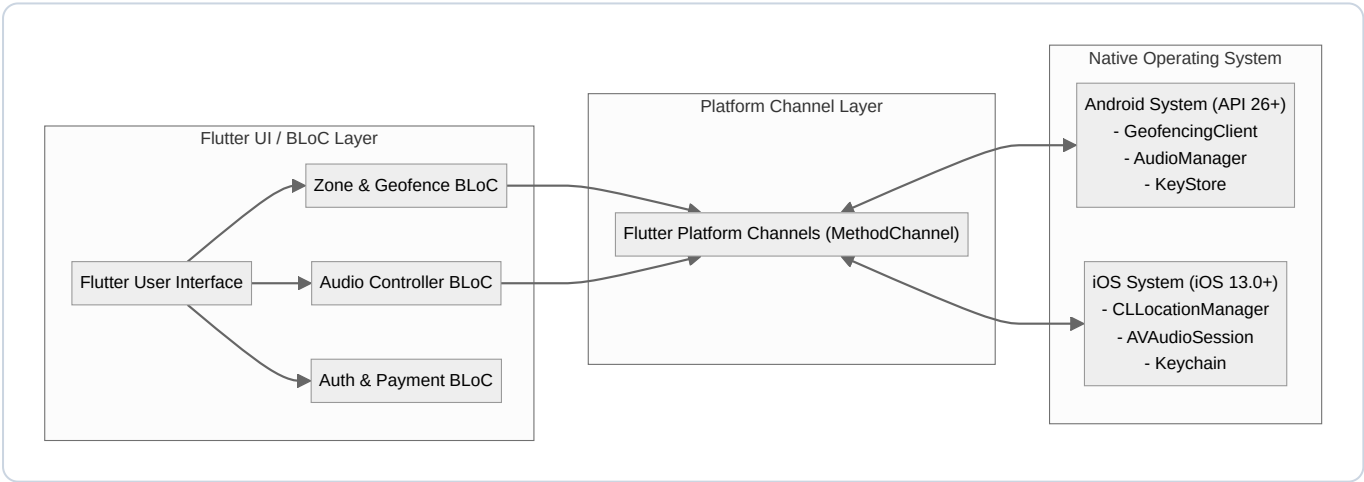
1.3 Definitions, Acronyms, and Abbreviations

Term	Full Definition
SRS	Software Requirements Specification (IEEE 830 / ISO 29148 Standard)
xdisturb	Project Code Identifier (<i>x-disrupt</i>)
OTP	One-Time Password (6-digit verification code sent via SMS)
JWT	JSON Web Token (Session authentication token)
Geofence	Virtual perimeter surrounding a physical geographic location
DND	Do Not Disturb (System sound profile mode)
Chapa SDK	Ethiopian payment aggregator gateway for Telebirr, CBE Birr, and credit cards
Hibir Audit	Security & quality evaluation platform (<code>audit.hibir.et</code>)

2. Overall System Description

2.1 Product Perspective

xdisturb functions as a background-resident, native mobile utility. It bridges high-level Flutter state management (BLoC pattern) with native Android and iOS hardware APIs through platform channels (`MethodChannel` and `EventChannel`).



2.2 User Classes & Personas

- 1. **Standard Mobile User:** Configures Silent Zones, relies on automated ringer profile switching, and uses free-tier features.
- 2. **Premium Subscriber:** Accesses unlimited custom geofences, advanced scheduler overrides, and processes subscription payments via Chapa.
- 3. **System Administrator:** Monitors system health, manages default place categories (Mosques, Churches, Libraries), and reviews Hibir Audit logs.

2.3 Operating Environment & Technical Dependencies

- **Mobile OS:** Android API Level 26+ (Android 8.0 Oreo) through Android 15; iOS 13.0 through iOS 18.
- **SDK & Language:** Flutter Framework with Dart SDK ^3.6.0 .
- **Hardware Permissions:**
 - Android: ACCESS_FINE_LOCATION , ACCESS_BACKGROUND_LOCATION , ACCESS_NOTIFICATION_POLICY (Do Not Disturb access), RECEIVE_BOOT_COMPLETED .
 - iOS: NSLocationAlwaysAndWhenInUseUsageDescription , NSLocationWhenInUseUsageDescription , Background Modes (location , processing).

3. System Functional Requirements

3.1 Module 1: Authentication & Session Management

FR-AUTH-01: Phone Number OTP Request & Verification

- **Description:** The system shall allow users to register and log in by providing their mobile phone number and verifying a 6-digit SMS One-Time Password (OTP).
- **Inputs:** Phone number (E.164 format, e.g., +251911223344), 6-digit numeric OTP code.
- **Processing:**
 1. User enters mobile number; app dispatches request to /api/v1/auth/request-otp .
 2. Server dispatches 6-digit SMS OTP code (valid for 300 seconds).
 3. User enters code; app posts to /api/v1/auth/verify-otp .
 4. Server validates OTP and returns JWT Access Token and Refresh Token.
- **Outputs:** Authenticated session context, JWT token payload.

FR-AUTH-02: Secure Storage Persistence

- **Description:** The system shall securely store JWT access tokens, refresh tokens, and user credentials on the mobile device using flutter_secure_storage .
- **Security Mechanics:**
 - Android: Encrypted using AES-256 via Android KeyStore (EncryptedSharedPreferences).
 - iOS: Encrypted using Apple Keychain services (kSecClassGenericPassword).

FR-AUTH-03: Referral Tracking & Promo Code Engine

- **Description:** The system shall track user referral counts, generate unique referral links, and apply promotional discount codes during subscription checkout.
- **Outputs:** Updated referral counter, applied discount percentage.

3.2 Module 2: Geofencing & Silent Zone Engine

FR-GEO-01: Continuous Background Geofence Monitoring

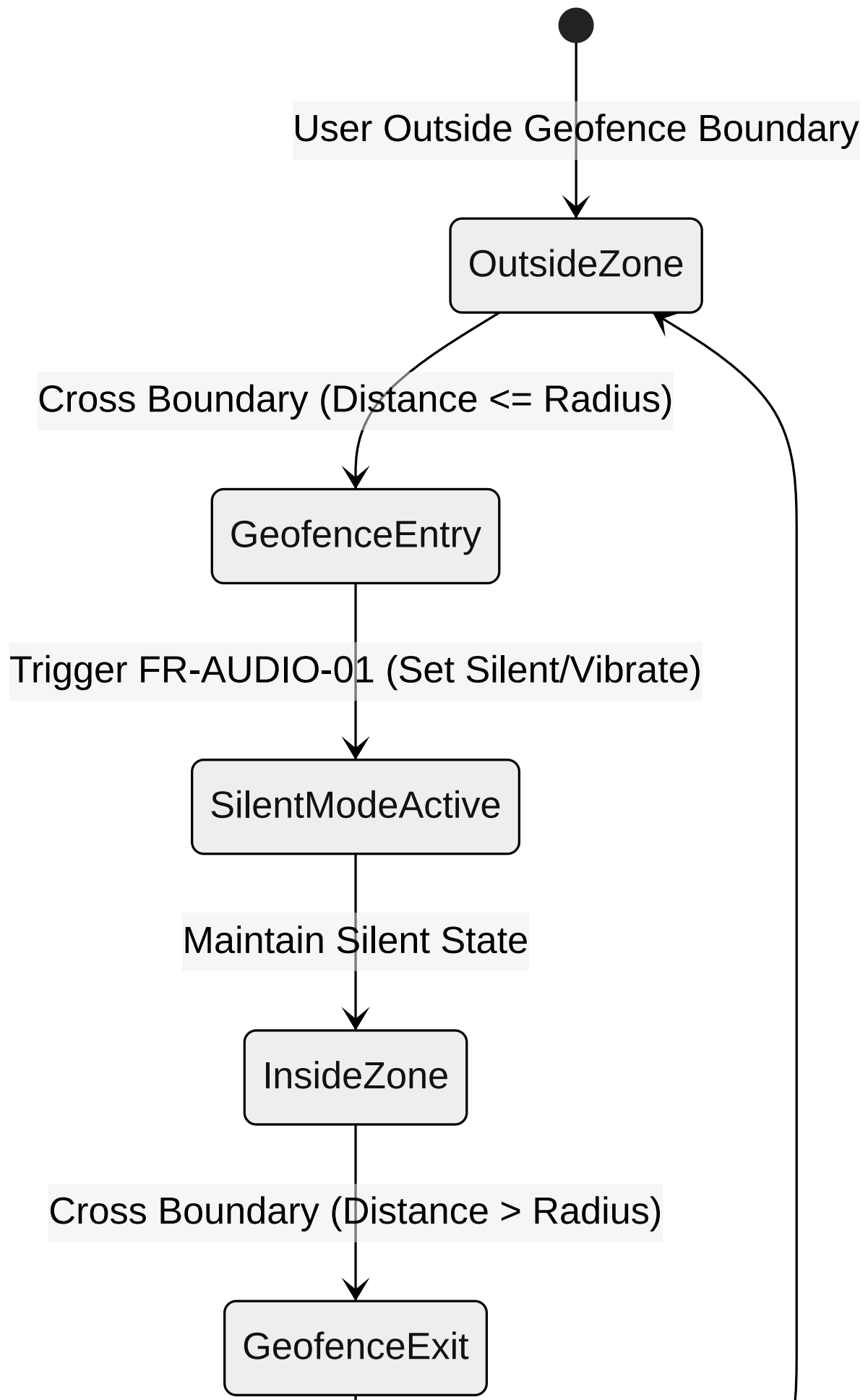
- **Description:** The system shall continuously monitor circular geofences with configurable radii between **50 meters and 500 meters** in background execution mode.
- **Performance Benchmark:** Background monitoring must survive OS battery optimization (Android Doze mode & iOS App Refresh) while maintaining location polling accuracy.

FR-GEO-02: Zone Categorization

- **Description:** The system shall organize Silent Zones into four distinct categories:
 1. **Mosque** (Default pre-populated religious zones)
 2. **Church** (Default pre-populated religious zones)
 3. **Library** (Default quiet learning zones)
 4. **Custom** (User-defined geographic perimeters)

FR-GEO-03: Interactive Map Marker Custom Zone Creation

- **Description:** The system shall allow users to create custom Silent Zones by dragging an interactive map marker to set latitude/longitude coordinates and adjusting a circular radius slider (50m – 500m).



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graph TD; Trigger[Trigger FR-AUDIO-02 (Restore Normal Ringer)] --> State[NormalModeRestored];
```

Trigger FR-AUDIO-02 (Restore Normal Ringer)

NormalModeRestored

3.3 Module 3: Automated Audio Profile Control

FR-AUDIO-01: Geofence ENTER Trigger & Audio Switching

- **Description:** Upon detecting a geofence `ENTER` event, the system shall automatically switch the device ringer mode to **Silent** or **Vibrate** according to user preference.
- **Native Call:**
 - Android: Invokes `AudioManager.setRingerMode(RINGER_MODE_SILENT)` or `RINGER_MODE_VIBRATE`. Requires Notification Policy access (`ACTION_NOTIFICATION_POLICY_ACCESS_SETTINGS`).
 - iOS: Triggers local notifications and audio override profiles via iOS System APIs.

FR-AUDIO-02: Geofence EXIT Trigger & Audio Restoration

- **Description:** Upon detecting a geofence `EXIT` event, the system shall automatically restore the device ringer mode to **Normal** (pre-entry ringer volume and profile state).

3.4 Module 4: Payments & Subscriptions

FR-PAY-01: Chapa SDK Integration for Mobile Payments

- **Description:** The system shall process subscription payments via the **Chapa Mobile SDK**, supporting local Ethiopian payment methods:
 - **Telebirr**
 - **CBE Birr**
 - **Bank Cards (Visa / Mastercard)**
- **Processing:**
 1. App initializes checkout session via `POST /api/v1/payments/initialize`.
 2. App opens Chapa payment sheet overlay.
 3. Upon payment completion, server receives webhook confirmation and unlocks Premium capabilities.

4. Non-Functional Requirements (NFR)

The non-functional specifications defined below adhere strictly to the target metrics established in the project charter:

Requirement ID	Category	Specification / Target Metric	Verification Method
NFR-PERF-01	Performance Latency	Geofence entry audio transition must occur within < 3.0 seconds of boundary crossing.	Field Testing & GPS Simulation
NFR-BATT-01	Battery Efficiency	Background geofence tracking must consume < 5% daily battery under standard usage.	Battery Historian / Xcode Energy Profiler
NFR-SEC-01	Transport Security	All API endpoints must enforce HTTPS (TLS 1.2/1.3) . Cleartext HTTP (<code>http://</code>) is strictly prohibited.	Network Packet Inspection
NFR-SEC-02	Storage Security	Session tokens and sensitive credentials must be encrypted in KeyStore / Keychain via <code>flutter_secure_storage</code> .	Static Code Analysis & Memory Dump
NFR-AUD-01	Hibir Audit Compliance	Application build must pass security evaluation guidelines specified by Hibir Audit (<code>audit.hibir.et</code>) .	Hibir Audit Certification

5. Requirements Traceability Matrix (RTM)

The matrix below maps each Functional and Non-Functional Requirement to its corresponding Flutter Component, Native Channel, and Test Case Reference:

Requirement ID	Description	Component / BLoC	Native Platform API / SDK	Test Case Reference
FR-AUTH-01	Phone OTP Verification	AuthBloc	Twilio / SMS Gateway API	TC-AUTH-001
FR-AUTH-02	JWT Secure Storage	StorageService	flutter_secure_storage (KeyStore/Keychain)	TC-AUTH-002
FR-AUTH-03	Referral & Promo Code	ReferralBloc	REST API (/api/v1/referrals)	TC-AUTH-003
FR-GE0-01	Background Geofence Tracking	LocationBloc	GeofencingClient (Android) / CLCircularRegion (iOS)	TC-GE0-001
FR-GE0-02	Zone Categorization	ZoneBloc	SQLite / Hive Local Database	TC-GE0-002
FR-GE0-03	Draggable Map Custom Zones	MapWidget	flutter_map / Google Maps API	TC-GE0-003
FR-AUDIO-01	Geofence ENTER Audio Switch	AudioBloc	AudioManager (Android) / AVAudioSession (iOS)	TC-AUDIO-001
FR-AUDIO-02	Geofence EXIT Audio Restore	AudioBloc	AudioManager (Android) / AVAudioSession (iOS)	TC-AUDIO-002
FR-PAY-01	Chapa Payment Processing	PaymentBloc	Chapa Flutter SDK (Telebirr, CBE Birr)	TC-PAY-001
NFR-PERF-01	Latency < 3.0s	System Core	GPS Geofence Hardware Listener	TC-NFR-001
NFR-BATT-01	Battery Drain < 5% / Day	Location Engine	Android Battery Historian	TC-NFR-002
NFR-SEC-01	TLS 1.2/1.3 Transport Security	ApiClient	Network Security Config / Dio Pinning	TC-NFR-003
NFR-SEC-02	Encrypted Credentials Storage	StorageService	Android KeyStore / iOS Keychain	TC-NFR-004
NFR-AUD-01	Hibir Audit Certification	Security Engine	audit.hibir.et Evaluation Suite	TC-NFR-005

Approval & Sign-Off:

- **Lead Mobile Engineer:** *Approved*
- **Hibir Audit Representative (audit.hibir.et):** *Approved*
- **System Architect:** *Approved*